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A palaeomagnetic investigation of a Permian redbed sequence from a mining drill core

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Summary. Samples spanning 66 m of Permian redbeds were collected from a vertical, but azimuthally unoriented, drill core obtained during a uranium exploration programme in the Lodève Basin of southern France. Inclinations are well grouped ($\bar{I} = -6.5°$, $k = 121.5$) and consistent with the known Permian palaeolatitude of Europe. Declinations have been estimated by using a sedimentological model involving tectonically-rotated cross-bed laminations. The results are very encouraging and suggest a pole at 151°E, 53°N ($\delta p = 1°$, $\delta m = 2°$, $N = 43$), which agrees well with the Eurasian apparent polar wander curve. The small polar discrepancy could well be due to secular variation, which is consistent with the small variance and presumed short time spanned by the sequence. Fully oriented surface samples confirm these findings, and support the validity of the orientation model. Temporal variations revealed by the stratigraphic ordering can be interpreted as slow dipole wobble, on a time-scale of 10^5 yr or more, on which is superimposed a more rapid non-dipole secular variation. It appears that the core stores of the world could become – at least under favourable circumstances – an important source of palaeomagnetic information.

Key words: palaeomagnetism, Permian, drill-core, cross-bedding, von Mises distribution

Introduction

There is a growing need for long sequences of stratigraphically ordered palaeomagnetic data in order to describe the temporal behaviour of the geomagnetic field. Over the last decade, investigations of recent lake sediments has led to considerable progress in this direction, but the bulk of the results available refer to the most recent 10^4 yr, or so, of geological time. Comparable data for earlier periods is much more sparse, and this prompted the present

attempt to extract useful palaeomagnetic data from a vertical, but azimuthally unoriented mining drill core. Apart from the notable exception of reversal stratigraphy studies on DSDP/IPOD cores, this type of material has attracted little attention; but even if the success rate turns out to be quite modest the core stores throughout the world offer a potential dataset well worth exploiting.

Procedure and results

The entire core is some 470 m long and was obtained during a uranium exploration programme in the Lodève Basin of southern France. This particular study involves a 66 m sequence of Permian redbeds which comprise the uppermost portion of the core. Two series of measurements are involved – magnetic and geometric. The first is straightforward and simply involves determining the remanence vector of each specimen relative to an arbitrary reference line scribed on the core. Specimens took the form of horizontal cylinders (25 mm diameter x 25 mm long) drilled perpendicular to the axis of the original 35 mm diameter core. Initial measurements made in the Montpellier laboratory employed a portable Molspin spinner magnetometer; the project was completed in the Edmonton laboratory using a Digico spinner magnetometer. The geometric measurements consisted of relating the arbitrary reference line to the stratification visible in each core segment, the intention being to relate this to the known bedding in the area to obtain an absolute reference frame. This was done by least-squares fitting of sinusoids to the stratification traces on the cylindrical surface of the original core. The actual data consisted of several – usually about 12 – distances between the selected stratification trace and an arbitrary horizontal circumference. Measurements were made with the aid of a graduated transparent plastic cylinder which fitted snugly over the core. This technique is applicable only to segments with simple stratification; zones of massive or contorted bedding were avoided. The sediments were deposited in a continental environment and exhibit abundant cross-bedding. This

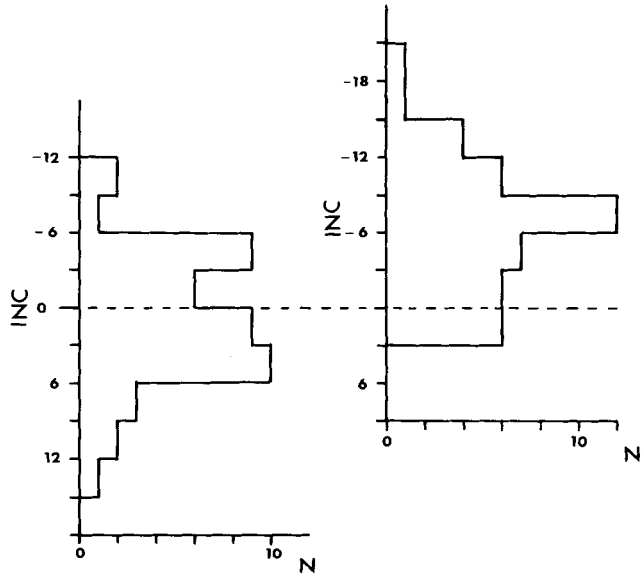


Figure 1. Inclination histograms ($N = 43$) before (left) and after (right) demagnetization treatment. The dashed line represents the horizontal.

complication obviously requires attention and we return to it further below when discussing the magnetic declinations. First, it is convenient to discuss the inclinations.

INCLINATION DATA

Fig. 1 illustrates the distribution of inclinations obtained from 43 specimens spread throughout the 66 m of core as evenly as circumstances allowed. These refer to the present reference frame; no bedding correction has been made. It is assumed that the initial verticality (attested by the driller's log) is retained throughout the 66 m involved. Forty of the cleaned results were obtained from thermal demagnetization to 500°C, the remaining three having been treated in a peak alternating field of 100 mT. Detailed pilot studies indicated that these treatments fall within the ranges where stable end-points were observed (Fig. 2). Seven of the thermally demagnetized samples were studied at the St Maur laboratory in Paris; we are very grateful to Brigitte Smith-Daignières for conducting these experiments on our behalf.

It is clear that there is a collective movement to more negative directions, which is consistent with the removal of a recent, steeply-inclined, secondary overprint. Following the

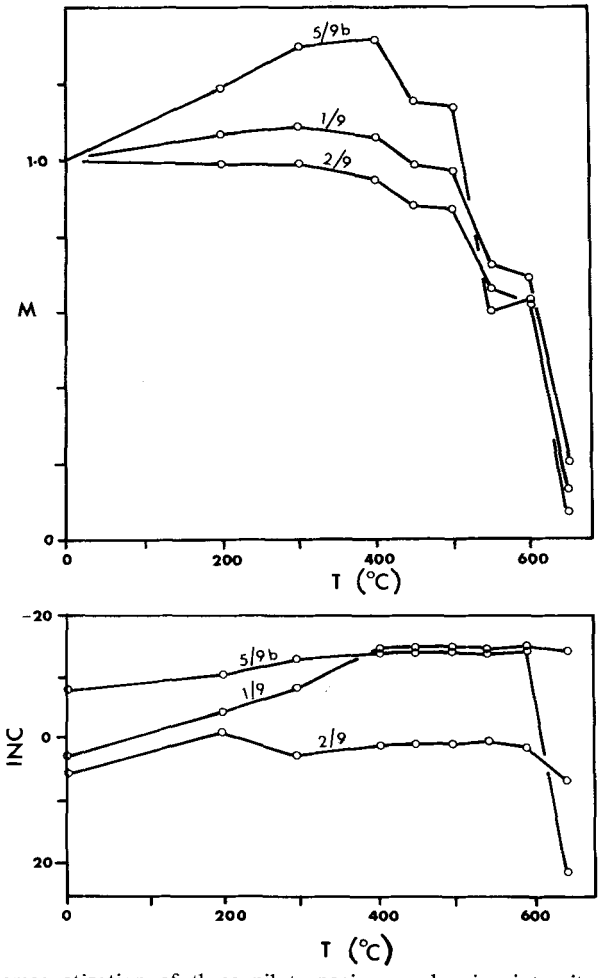


Figure 2. Thermal demagnetization of three pilot specimens, showing intensity changes (above) and inclination changes (below).

procedure of McFadden & Reid (1982) we obtain a mean inclination of -6.5° with a Fisher precision-parameter (k) of 121.5.

The mean inclination cannot be directly converted to a palaeolatitude because the strata are known to dip at 10–20°; azimuthal control is, therefore, necessary before a dip correction can be applied. For this step we turn to the following attempt to extract declination information from this material.

DECLINATION DATA

These sediments were formed in an intermittently fluvial sub-desert environment, and cross-bedding is consequently ubiquitous. Following a detailed study of the sedimentology, Kruseman (1962) concludes that the direction of transport was from west to east. Potter & Pettijohn (1977) have made an extensive survey of published data and conclude that averages for the dip of cross-bedding generally fall between 18 and 25°. We combine these conclusions and propose that, on average, the original cross-bed laminations inclined 20° to the East. The geological map of the area, and our own measurements in the vicinity of the drill hole, indicate a tectonic dip of 16° towards N165°E. Tilting the cross-bed mode axis appropriately leads to a maximum-likelihood strike of N211°E (see Fig. 3a). We assign this value to the stratification in each measured core segment and obtain the distribution of magnetic declinations shown in Fig. 3(b).

Since the magnetic vectors are close to the horizontal we adopt a circular normal (von Mises) distribution to analyse the declinations obtained by the above procedure. This yields $\bar{D} = 197.2^\circ$ with a standard deviation of 28.4°. Assuming $k = 121.5$, as deduced from the analysis of inclinations, one expects an angular standard deviation ($81/\sqrt{k}$) of 7.3°; the excess reflects real variation in cross-bed orientation. The amount of azimuthal variation actually observed in different cases depends on the particular environment, but inspection

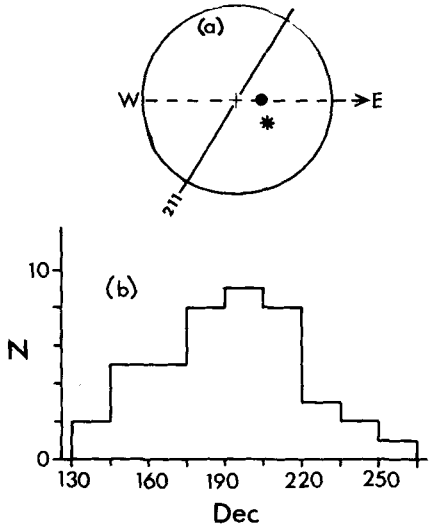


Figure 3. (a) Sketch of the orientation procedure. The transport direction is from west to east and the dip is 20° (the pole to the average cross-bed laminations is represented by a dot on the upper hemisphere). After tectonic dip correction (16° towards N165°E) this pole moves to the position shown by the star, which corresponds to a strike of N211°E. (b) Histogram of 43 magnetic declinations assuming the model set out in (a).

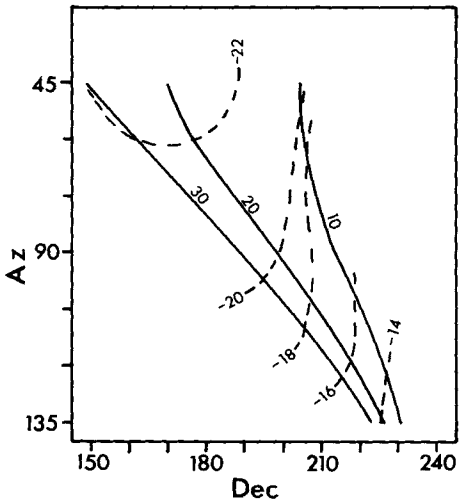


Figure 4. Nomogram illustrating the consequences of various sedimentological models on the final mean magnetic vector. The ordinate (Az) represents the transport direction in degrees clockwise from north; the abscissa gives the mean magnetic declination (after tectonic dip correction). Continuous curves show the effect of different average cross-bed dips and dashed curves the resulting mean magnetic inclination (after tectonic dip correction).

vary from N45°E to N135°E, and the cross-bedding dip from 10° to 30° – it is unlikely that the average falls outside these generous bounds. One sees that the final dip-corrected inclination falls between −13° and −23°, corresponding to palaeolatitude limits of ~7° to ~12°. It is not possible to obtain inclinations as low as that found by Kruseman or that from our own surface samples. However, it is noteworthy that a combination of an average transport direction of N90°E (Kruseman 1962) and an average cross-bedding dip in the range 18° to 25° (Potter & Pettijohn 1977) yields a mean declination close to 200°, which is in good agreement with both earlier datasets. We therefore adopt the overall dip-corrected mean

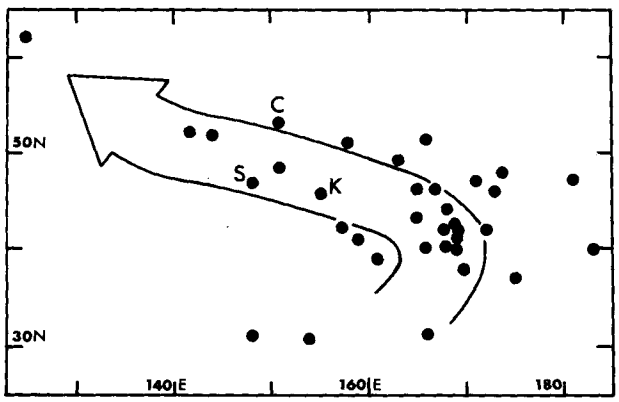


Figure 5. Permian palaeomagnetic poles and apparent polar wander for Eurasia (after Irving & Irving 1982). There are 33 poles, including that marked *K* which is the Saxonian pole of Kruseman (1962). Virtual geomagnetic poles from the present study are labelled *S* (surface samples) and *C* (drill core samples). The polar path is schematically represented as a band about 10° wide, which is typical of the errors on the moving averages obtained by Irving & Irving (1982 – see their fig. 4 and table IV).

direction of magnetization given above, namely $\bar{D} = 199°$, $\bar{I} = -20°$, $k = 122$ for the 66 m sequence of strata.

Poles corresponding to the drill-core samples (151°E, 53°N, $\delta p = 1°$, $\delta m = 2°$, $N = 43$), the surface samples (148°E, 47°N, $\delta p = 2°$, $\delta m = 4°$, $N = 15$), and Kruseman's Saxonian data are plotted in Fig. 5, where they are compared with the Eurasian apparent polar wander curve proposed by Irving & Irving (1982). Agreement is good. The differences observed could well be due to secular variation which is discussed more fully in the next section. For the moment we reiterate that Kruseman's result represents an average over some 1500 m whereas our data represent only 66 and 10 m for the drill-core and surface samples respectively. Moreover, our two datasets may be distinct in time since there is a mapped fault running between the two sampling localities. It is also possible that Kruseman's average is slightly offset by the failure to completely remove recent overprints from all his selected samples. Be that as it may, it appears that reliable palaeomagnetic results can be obtained from unoriented cores – at least when circumstances are favourable.

Palaeosecular variation

The inclination analysis gives $k = 121.5$ with a corresponding angular dispersion of 7.3° – well below the equatorial Permian value of about 15° given by the global summary of Irving

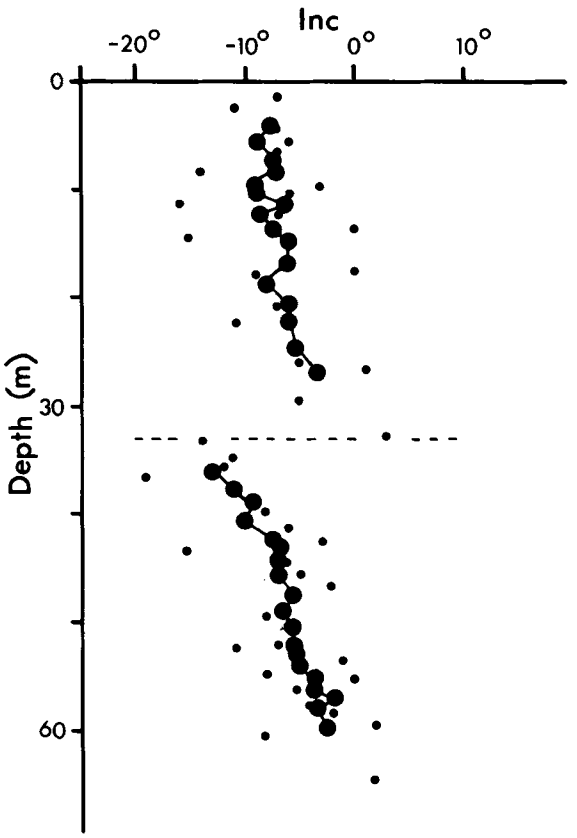


Figure 6. Inclination values (before tectonic dip correction) versus depth in the core. The two continuous lines represent 5-point moving averages above and below a proposed hiatus (shown by the dashed line) which is marked by a coarse gravel layer.

& Pullaiah (1976). This implies that insufficient time is represented to average out the temporal variations of the field and raises the question of the origin of the variations observed.

To investigate the temporal variation of the geomagnetic field which occurred during deposition of this sedimentary sequence one requires declination and inclination for each horizon. This is not possible here because the attitude of the cross-bedding at each individual horizon is unknown. Such success as the above orientation procedure enjoys results from the use of averages, not individual data. Nevertheless, a pattern in the uncorrected inclination data is clearly apparent (Fig. 6). The results fall into two distinct clouds with an abrupt break near a depth of 32 m. Subsequently, the drillers' log was obtained and it was gratifying to discover that there is a 50 cm coarse gravel at this level. Furthermore, it is the only such gravel recorded in the entire 66 m of sediment. It presumably reflects a hiatus of unknown duration. The inclination trends above and below this break may be due to dipole wobble, with the higher frequency content then being attributed to secular variation of the non-dipole field. The smoothed curves obtained from a 5-point moving average indicate the possible form of the dipole wobble. Observed deviations up to 10° then represent non-dipole effects – certainly an acceptable magnitude for such perturbations. Although a pattern emerges when viewed over tens of metres, there is very little sequential ordering on a scale of 1 or 2 m. This suggests that the sample spacing is too coarse to yield smooth secular variation paths. If these took place on a time-scale of $\sim 10^3$ yr, as is the case for the recent past, then

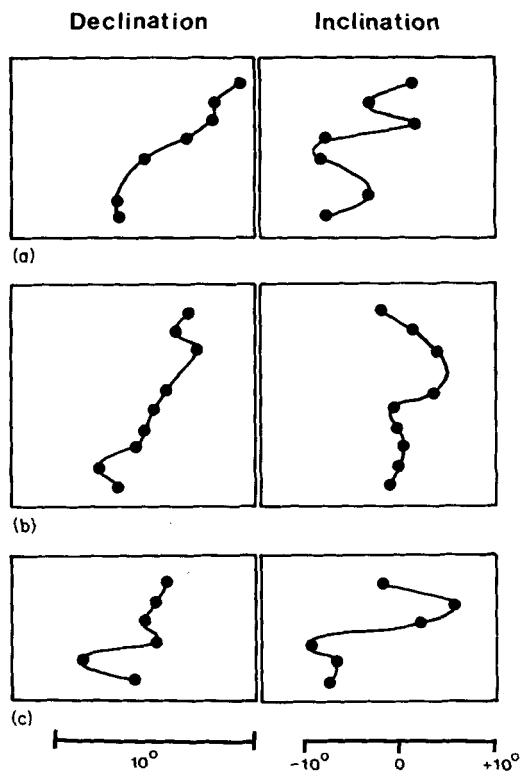


Figure 7. Inclination and relative declination versus depth in three individual core segments near depths of 1 m (a), 40 m (b), and 46 m (c). The depth ranges are 37 cm (a), 46 cm (b), and 28 cm (c). Declination origin is arbitrary.

we can perhaps associate a time span of this order with our typical sampling interval; this in turn leads to an order-of-magnitude estimate of 1 mm yr^{-1} for the deposition rate. Each 30 m section would then represent some 3×10^4 yr, which would imply a time scale of $\gtrsim 10^5$ yr for dipole wobble. Three of the longer core segments were investigated in more detail to test this general interpretation. As with the other core segments these cannot be azimuthally oriented *a priori*, but relative orientation between individual specimens is possible. As can be seen in Fig. 7, successive vectors do fall on smooth curves, and this supports the idea that the high-frequency scatter does represent a real geomagnetic phenomenon.

Van den Ende (1977) reaches very similar conclusions from his study of fully-orientated samples of Permian redbeds from the Dôme de Barrot which is situated some 300 km to the East of the Lodève Basin. A detailed study of two particular sequences (1.3 and 2.3 m thick) reveals field-vector trajectories that sweep out open loops which are very reminiscent of the recent secular variation. Exploiting this analogy, van den Ende deduces a sedimentation rate of 1 mm yr^{-1}. A more widely-sample section (26 horizons spanning 38 m, four samples per horizon) suggests longer-period variations which he interprets as dipole wobble on a time scale of $1-5 \times 10^4$ yr. Without adequate dating control all these conclusions – both our own and those of van den Ende – must be regarded as rather tentative, but the nature of these sedimentary sequences makes it unlikely that the absolute chronology will be significantly improved in the near future. Nevertheless, it does seem that the gross properties of the geomagnetic field in the deep geological past is recoverable and that appropriate enquiry will pay dividends.

Acknowledgments

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